T.C.
GEBZE TECHNICAL UNIVERSITY
PHYSICS DEPARTMENT

PHYSICS LABORATORY I
EXPERIMENT REPORT

THE NAME OF THE EXPERIMENT

Hooke's Law

GEBZE TEKNİK ÜNİVERSİTESİ

PREPARED BY

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690 - 489 = 210,1 cm
69,0 - 49,7 = 19,3 cm

Experimental Procedure:

The experimental set-up to measure the spring constants is shown in Fig.10.1.

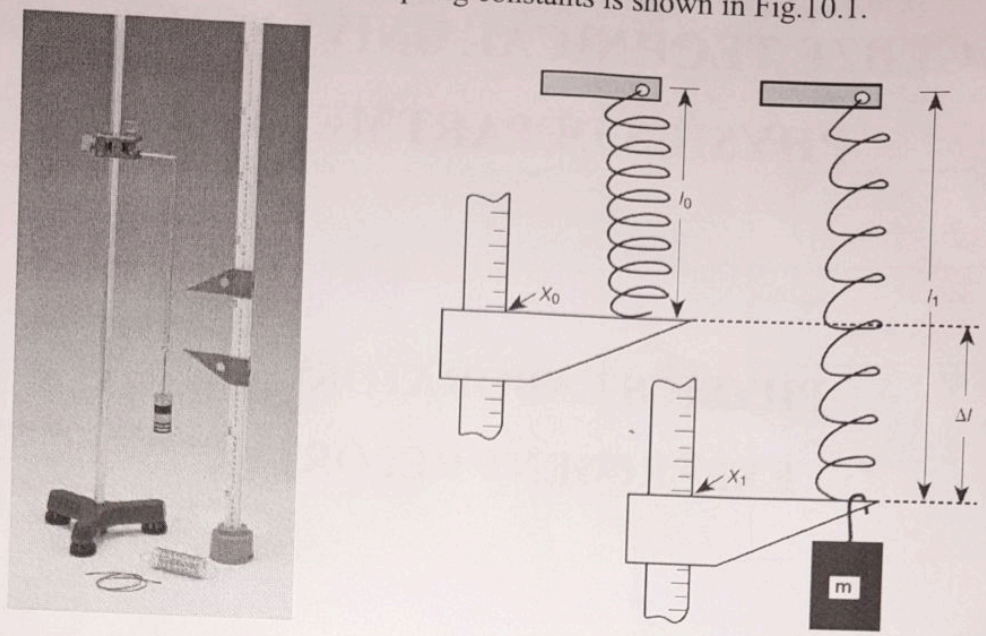


Figure 10.1: Experimental set-up: Hooke's law.

Hook's Law:

1. Measure the initial equilibrium position x_0 of each spring (thin and thick) and mass of the holder.

$x_0^{thin} =$ 20,1 (cm) $x_0^{thick} =$ 19,3 (cm) $m_{holder} =$ 10 (gr)

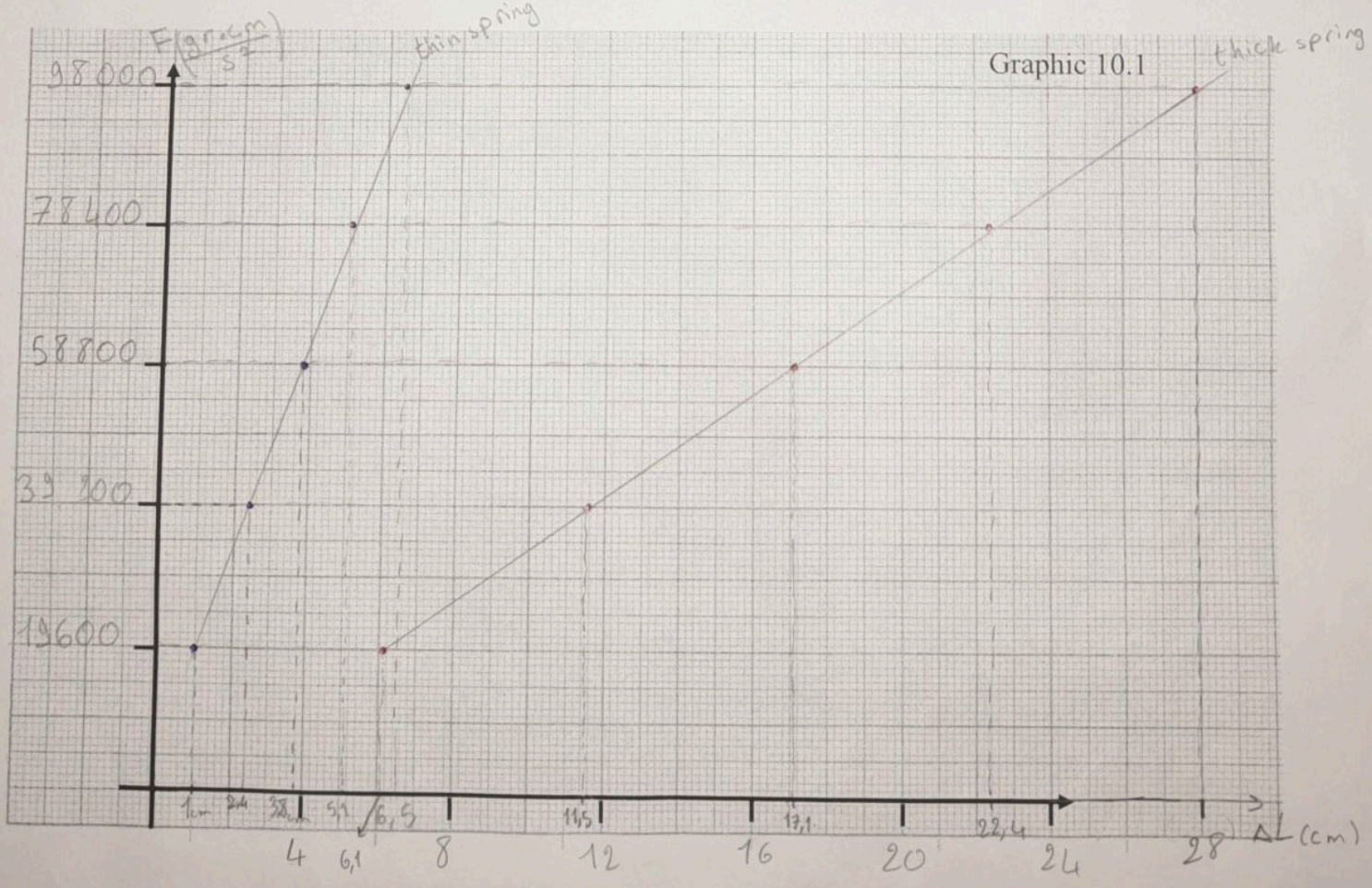
2. Suspend a mass on the holder, then measure the displacements from the equilibrium position for each spring. Don't forget to use the total mass (additional mass + mass of the holder) attached to the spring in the calculations.
3. One after another, suspend an additional mass by 20 gr increments to a total of 100 gr and read the corresponding equilibrium position x_i, then calculate the change of length ΔL. Record the values in Table 10.1
4. Calculate the weight (force) $F = mg$ (g=980 cm/s^2) and also note these values in Table 10.1.

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Table 10.1: Spring length L as a function of the suspended weights.

m (gr)	$F=mg$ (gr.cm/s^2)	x_i^{thin} (cm)	$\Delta L_{thin} = x_i^{thin} - x_0^{thin}$ (cm)	x_i^{thick} (cm)	$\Delta L_{thick} = x_i^{thick} - x_0^{thick}$ (cm)
20	20.980 = 19600	69,0 - 47,9 = 21,1	ΔL = 21,1 - 20,1 = 1cm	69,0 - 43,6 = 25,4	ΔL = 25,4 - 19,3 = 6,1 cm
40	40.980 = 39200	69,0 - 46,5 = 22,5	ΔL = 22,5 - 20,1 = 2,4 cm	69,0 - 38,2 = 30,8	ΔL = 30,8 - 19,3 = 11,5cm
60	60.980 = 58800	69,0 - 45,1 = 23,9	ΔL = 23,9 - 20,1 = 3,8 cm	69,0 - 32,6 = 36,4	ΔL = 36,4 - 19,3 = 17,1 cm
80	80.980 = 78400	69,0 - 43,8 = 25,2	ΔL = 25,2 - 20,1 = 5,1cm	69,0 - 27,3 = 41,7	ΔL = 41,7 - 19,3 = 22,4cm
100	100.980 = 98000	69,0 - 42,4 = 26,6	ΔL = 26,6 - 20,1 = 6,5 cm	69,0 - 21,7 = 47,3	ΔL = 47,3 - 19,3 = 28,0 cm

Use the values in Table 10.1 and plot F-ΔL graphs of each spring on reserved millimetric space as x-axis *the change of length* (ΔL) and y-axis the force (F). Represent the values in the table as points on your graph.



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If we take into account our theoretical considerations, we expect a line passing through those points. The Eq. 10.7 $F = -k\Delta L$ describes a linear ($y=kx$) relation between the force F acting on the spring and the change of length ΔL, with the slope spring constant k . Use the slope k and, which will be calculated in the following step, plot $y=kx$ line on your graph. Observe the fitness of the line to your data points.

You are expected to calculate the slope k. The slope of the line could be calculated using the values in Table 10.1 with the statistical fitting method called "*least squares method*".

A) Thin spring;

Calculate two terms that will be used in the equations below.

$$\sum_{i=1}^{5} \Delta L_i F_i = L_1 \cdot F_1 + L_2 \cdot F_2 + L_3 \cdot F_3 + L_4 \cdot F_4 + L_5 \cdot F_5$$

$$= 1 \cdot 19600 + 2,4 \cdot 39200 + 3,8 \cdot 58800 + 5,1 \cdot 78400 + 6,5 \cdot 98000$$

$$= 1\,373\,960 \ \frac{gr.cm^2}{s^2}$$

$$\sum_{i=1}^{5} \Delta L_i^2 = L_1^2 + L_2^2 + L_3^2 + L_4^2 + L_5^2 = 1^2 + (2,4)^2 + (3,8)^2 + (5,1)^2 + (6,5)^2$$

$$= 89,46 \ cm^2$$

Substitute those values in equation and calculate the slope (spring constant) k_{thin}.

$$k_{thin} = \frac{\sum_{i=1}^{5} \Delta L_i F_i}{\sum_{i=1}^{5} \Delta L_i^2} = \frac{1\,373\,960 \ \frac{gr.cm^2}{s^2}}{89,46 \ cm^2} = 15\,358,37246 \ gr/s^2$$

Thin spring constant k_{thin}= 15358,37246 ~~gr/s^2~~)

B) Thick spring;

Calculate two terms that will be used in the equations below.

$$\sum_{i=1}^{5} \Delta L_i F_i = L_1 \cdot F_1 + L_2 \cdot F_2 + L_3 \cdot F_3 + L_4 \cdot F_4 + L_5 \cdot F_5$$

$$= 6,1 \cdot 19600 + 11,5 \cdot 39200 + 17,1 \cdot 58800 + 22,4 \cdot 78400 + 28,0 \cdot 98000$$

$$= 105089320 \ \frac{gr.cm^2}{s^2}$$

$$\sum_{i=1}^{5} \Delta L_i^2 = L_1^2 + L_2^2 + L_3^2 + L_4^2 + L_5^2$$

$$= 1747,63 \ cm^2$$

Substitute those values in the equation and calculate the slope (spring constant) k_{thick}

$$k_{thick} = \frac{\sum_{i=1}^{5} \Delta L_i F_i}{\sum_{i=1}^{5} \Delta L_i^2} = \frac{105089320 \ \frac{gr.cm^2}{s^2}}{1747,63 \ cm^2} = 60132,47655 \ gr/s^2$$

Thick spring constant k_{thick}= 60132,47655 ~~gr/s^2~~)

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Harmonic oscillation:

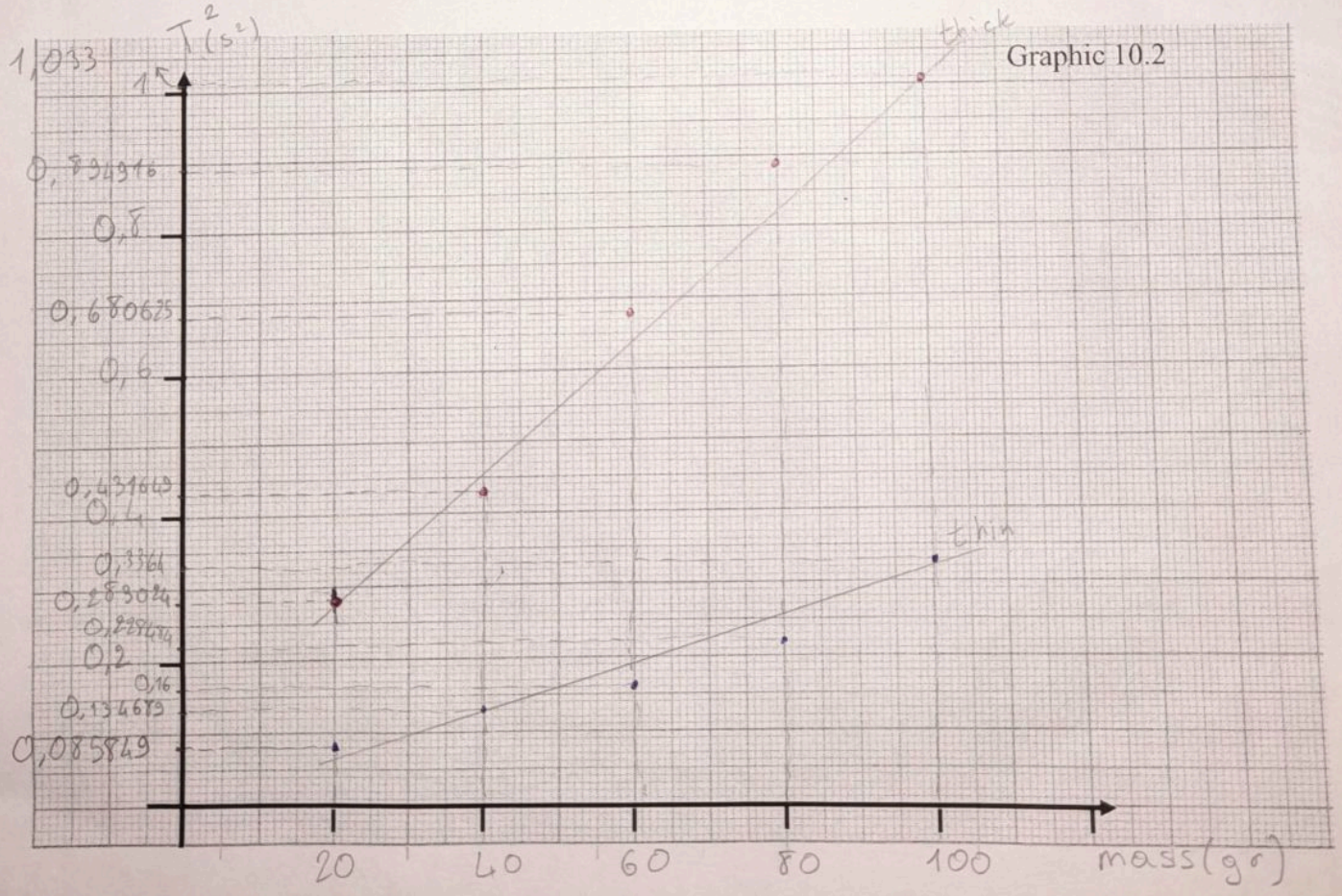
1. In this step, suspend the same masses on each spring in turn. Pull the mass little bit down then release. Measure the time of the oscillation of the spring to complete 10 cycles for each mass by using the stopwatch. Divide each time by 10 to find the time for one period for each mass and record the values in table 10.2.

2. Calculate the period of oscillation T and square of the period of oscillation T^2 and fill in Table 10.2.

Table 10.2: Period of oscillation T as function of the suspended mass.

m (gr)	Δt_{thin} (s)	$T_{thin} = \frac{\Delta t_{thin}}{10}$ (s)	T^2_{thin} (s^2)	Δt_{thick} (s)	$T_{thick} = \frac{\Delta t_{thick}}{10}$ (s)	T^2_{thick} (s^2)
20	2,93 s	0,293 s	0,085849	5,32 s	0,532 s	0,283024
40	3,67 s	0,367 s	0,134689	6,57 s	0,657 s	0,431649
60	4,00 s	0,400 s	0,160	8,25 s	0,825 s	0,680625
80	4,78 s	0,478 s	0,228484	9,46 s	0,946 s	0,894916
100	5,80 s	0,580 s	0,3364	10,33 s	1,033 s	1,067089

Graphic 10.2



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Use the values in the Table 10.2 and plot T^2-m graph for both spring on the same graph with x-axis *the mass* (m) and y-axis *square of the period of oscillation* (T^2). Represent the values in the table 10.2 as points on your graph. If one takes squares of both sides of the Eq. 10.7, $T^2 = \frac{4\pi^2}{k} m$ is obtained and this equation describes a linear ($y=ax$) relation between the square of the period of oscillation T^2 and the mass m, with the slope $a = \frac{4\pi^2}{k}$. Use the slope a and, which will be calculated in the following step, plot $y=ax$ line on your graph. Observe the fitness of the line to your data points.

You are expected to calculate the slope $a = \frac{4\pi^2}{k}$. The slope of the line could be calculated using the values in the table 10.2 with the statistical fitting method called "*least squares method*".

A) Thin spring;

Calculate the two terms that will be used in the equations below.

$$\sum_{i=1}^{5} m_i T_i^2 = m_1 \cdot T_1^2 + m_2 T_2^2 + m_3 T_3^2 + m_4 T_4^2 + m_5 T_5^2$$

$$= 20.0,005849 + 40 \cdot 0,134689 + 60 \cdot 0,16 + 80 \cdot 0,228474 + 100 \cdot 0,3364$$

$$= 68,62326 \text{ gr} \cdot s^2$$

$$\sum_{i=1}^{5} m_i^2 = (m_1)^2 + (m_2)^2 + (m_3)^2 + (m_4)^2 + (m_5)^2$$

$$= 20^2 + 40^2 + 60^2 + 80^2 + 100^2 = 22000 \text{ gr}^2$$

Substitute those values in equation below and calculate the slope a.

$$a = \frac{\sum_{i=1}^{5} m_i T_i^2}{\sum_{i=1}^{5} m_i^2} = \frac{68,62326 \text{ gr} \cdot s^2}{22000 \text{ gr}^2} = 0,003119239 \ s^2/\text{gr}$$

The spring constant k_{thin} can be calculated from the slope a according to

$$k_{thin} = \frac{4\pi^2}{a} = \frac{4\pi^2}{0,003119239 \ s^2/\text{gr}} = 12656.428(0029) \text{ gr}/s^2$$

B) Thick spring;

Calculate the two terms that will be used in the equations below.

$$\sum_{i=1}^{5} m_i T_i^2 = m_1 \cdot T_1^2 + m_2 T_2^2 + m_3 \cdot T_3^2 + m_4 \cdot T_4^2 + m_5 \cdot T_5^2$$

$$= 20 \cdot 0,532 + 40 \cdot 0,657 + 60 \cdot 0,825 + 80 \cdot 0,946 + 100 \cdot 1,070 89$$

$$= 269,189 \text{ gr} \cdot s^2$$

$$\sum_{i=1}^{5} m_i^2 = (m_1)^2 + (m_2)^2 + (m_3)^2 + (m_4)^2 + (m_5)^2$$

$$= 20^2 + 40^2 + 60^2 + 80^2 + 100^2 = 22000 \text{ gr}^2$$

Substitute those values in equation below and calculate the slope a.

$$a = \frac{\sum_{i=1}^{5} m_i T_i^2}{\sum_{i=1}^{5} m_i^2} = \frac{269,189 \text{ gr} \cdot s^2}{22000 \text{ gr}^2} = 0,012235863 \ s^2/\text{gr}$$

The spring constant k_{thick} can be calculated from the slope a according to

$$k_{thick} = \frac{4\pi^2}{a} = \frac{4\pi^2}{0,012235863 \ s^2/\text{gr}} = 3226.4514(2434) \text{ gr}/s^2$$

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Compare the spring constants k_{thin} and k_{thick} values calculated by using the Hook's law and harmonic oscillation with each other. Discuss the reasons for probable differences.

When I hang the same mass on two helical springs with different spring constants their amount of elengation change. And we saw that the thick spring streches more than the thin spring.

Conclusion, Comment and Discussion:

(Tips: Give detailed explanations about what you've learned in the experiment and also explain the possible errors and their reasons.)

I learned in this experiment that the gravitational force is inversely proportional with the spring constant and directly with the amount of elengation, $F=-k\Delta x$
I learned that the period is directy proportional with mass and is inverly with spring constan in simple harmonic motion because $T=2\pi\sqrt{\frac{m}{k}}$
The thin spring period is small than the period of the thick spring when I hang the same mass, And thick spring streches more than the thine spring when i hang the same mass.
I learn how to measure period and draw graphic.

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Questions:

9,763 m/s² 9,798 m/s²

1) When you measure the period of the same spring at *the poles* and near *the equator*, do you observe any difference? How do you explain this?

$T = 2\pi\sqrt{\frac{m}{g}}$

The period in the equator is longest than the period of the same spring in poles because gravity in poles: 9,763 m/s² and in equator = 9,798 m/s because the period direcllypropotion with mass and is propotional with gravity.

2) A mass *m* is hanged to a 10 cm spring with a spring constant *k* and the system performs a simple harmonic motion. Then the spring is divided into two pieces as 3 cm and 7 cm. The same *m* mass is hanged to both springs. How do the period and the spring constants change? Please explain.

The spring constant of both springs increase and both springs getstipper. But the spring with 3cm lengh will be more stipper than the 7cm's spring and the 3cm's spring will have a bigger spring constant value than the 7cm's spring. The period of both spoings decrase. But the period value of the 3cm's spring will be lesser than the period value of the 7cm's spring.

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